Supplementary Results: When Does INT8 Compression Preserve Federated Intrusion Detection on Microcontrollers?

Every table is generated by `scripts/build_supplement.py` from records that already exist in the repository (no retraining, no new statistical tests); the complete rows, with file names and SHA-256 of every export, are in the CSV files named in each caption (`paper/supplement/`). Recall and FAR in %; "test FAR" is the false-alarm rate *achieved* on the test split at the threshold chosen for the stated *validation* FAR budget. Reported rows use the per-tensor exports; first-selected per-channel exports are kept in S1 and marked as superseded.

Table 1: S8 (`S8_deployable_models.csv`): principal models, 80 KB arms; mean over three runs. Latency: median invoke on the device model of the same arm (seed 1). Minutes: federated training wall-clock per run on one PC.

split	arm	recall@1%	test FAR	FAR range	recall@5%	test FAR	S3 ms	ESP32 ms	training	teacher min	compact min
TON	F	96.68 ± 1.38	0.84	0.66-1.18	99.86	4.14			teacher (60 rounds)	3.4	
TON	F8	98.35 ± 0.06	1.19	1.12-1.27	99.85	4.17			teacher (60 rounds) + PTQ	3.4	
TON	P	96.57 ± 1.89	0.98	0.95-1.00	99.74	4.08	5.55	13.18	teacher (60 rounds) + 3 fine-tuning rounds + PTQ	3.4	
TON	QC	97.86 ± 0.44	1.01	0.91-1.06	99.83	4.15	5.55	13.2	teacher (60) + 3 fine-tuning + 2 QAT rounds	3.4	
TON	R	98.51 ± 0.11	1.16	1.10-1.27	99.81	3.98	5.56	13.19	teacher (60) + 3 distilled fine-tuning (teacher pass) + 2 QAT rounds	3.4	
TON	S80	97.57 ± 0.52	1.01	0.93-1.16	99.83	4.23	5.56	13.35	compact model (60 rounds) + PTQ	3.4	2.7
TON	SQC	97.71 ± 1.09	1.01	0.91-1.14	99.81	3.91			compact model (60) + 2 QAT rounds	3.4	2.7
CIC	F	96.89 ± 2.05	0.98	0.98-0.99	99.89	5.07			teacher (60 rounds)	40.3	
CIC	F8	99.33 ± 0.54	0.9	0.83-0.94	99.97	4.94			teacher (60 rounds) + PTQ	40.3	
CIC	P	99.12 ± 0.37	0.92	0.81-0.98	99.81	4.56	6.47	15.36	teacher (60 rounds) + 3 fine-tuning rounds + PTQ	40.3	
CIC	QC	99.57 ± 0.1	0.94	0.92-0.96	99.81	3.79	6.48	15.27	teacher (60) + 3 fine-tuning + 2 QAT rounds	40.3	
CIC	R	99.6 ± 0.15	0.92	0.84-0.95	99.95	4.99	6.48	15.38	teacher (60) + 3 distilled fine-tuning (teacher pass) + 2 QAT rounds	40.3	
CIC	S80	99.57 ± 0.24	0.97	0.96-0.99	99.97	4.95	6.47	15.37	compact model (60 rounds) + PTQ	40.3	25.9
CIC	SQC	99.59 ± 0.07	0.97	0.95-0.98	99.83	3.84			compact model (60) + 2 QAT rounds	40.3	25.9
TON-hosts	F	98.2 ± 0.11	3.18	0.00-4.77	99.91	4.05			teacher (60 rounds)	3.2	
TON-hosts	F8	98.14 ± 0.17	0.0	0.00-0.00	99.78	2.18			teacher (60 rounds) + PTQ	3.2	
TON-hosts	P	93.81 ± 0.27	0.0	0.00-0.00	99.7	3.99			teacher (60 rounds) + 3 fine-tuning rounds + PTQ	3.2	
TON-hosts	QC	96.3 ± 0.87	0.0	0.00-0.00	99.81	2.48			teacher (60) + 3 fine-tuning + 2 QAT rounds	3.2	
TON-hosts	R	96.57 ± 0.62	0.0	0.00-0.00	99.81	2.14			teacher (60) + 3 distilled fine-tuning (teacher pass) + 2 QAT rounds	3.2	
TON-hosts	S80	96.48 ± 0.94	0.0	0.00-0.00	99.81	3.88			compact model (60 rounds) + PTQ	3.2	2.6
TON-hosts	SQC	95.96 ± 1.6	0.0	0.00-0.00	99.79	3.31			compact model (60) + 2 QAT rounds	3.2	2.6

Table 2: S1 (`S1_operating_points.csv`): recall / achieved test FAR per run (seed 1, 2, 3) at validation FAR budgets of 1% and 5%, 80 KB arms and the teacher. 128 KB arms and one-client fine-tuning are in the CSV.

split	arm	1%: s1	s2	s3	5%: s1	s2	s3
TON	F	96.00 / 0.68	95.77 / 0.66	98.27 / 1.18	99.87 / 4.34	99.88 / 4.05	99.85 / 4.02
TON	F8	98.28 / 1.12	98.40 / 1.27	98.38 / 1.18	99.86 / 4.38	99.86 / 4.11	99.82 / 4.02
TON	P	97.98 / 1.00	94.42 / 0.95	97.32 / 1.00	99.78 / 4.11	99.75 / 4.13	99.69 / 4.00
TON	Q	74.43 / 0.83	0.00 / 0.00	91.86 / 1.00	98.64 / 4.27	98.91 / 3.80	98.72 / 4.48

split	arm	1%: s1	s2	s3	5%: s1	s2	s3
TON	QC	98.37 / 0.91	97.68 / 1.06	97.54 / 1.06	99.85 / 4.11	99.83 / 4.23	99.80 / 4.11
TON	R	98.41 / 1.12	98.50 / 1.10	98.62 / 1.27	99.85 / 3.98	99.83 / 4.32	99.77 / 3.65
TON	S80	97.67 / 0.93	98.02 / 0.93	97.00 / 1.16	99.83 / 4.23	99.80 / 4.38	99.85 / 4.09
TON	SQ	90.31 / 0.93	0.00 / 0.00	86.63 / 0.73	98.59 / 3.92	98.94 / 4.19	99.51 / 4.75
TON	SQC	96.46 / 0.91	98.29 / 0.98	98.40 / 1.14	99.80 / 3.51	99.82 / 4.05	99.82 / 4.17
CIC	F	97.82 / 0.98	94.54 / 0.98	98.32 / 0.99	99.97 / 5.11	99.72 / 5.07	99.98 / 5.03
CIC	F8	99.71 / 0.92	99.57 / 0.94	98.72 / 0.83	99.98 / 5.00	99.97 / 5.04	99.96 / 4.77
CIC	P	98.87 / 0.98	98.95 / 0.81	99.55 / 0.95	99.84 / 4.57	99.84 / 4.80	99.74 / 4.30
CIC	Q	83.39 / 0.93	90.32 / 0.93	72.05 / 0.54	96.27 / 4.83	99.36 / 4.74	95.29 / 4.82
CIC	QC	99.62 / 0.93	99.45 / 0.92	99.63 / 0.96	99.80 / 3.99	99.82 / 4.23	99.80 / 3.16
CIC	R	99.70 / 0.95	99.43 / 0.84	99.68 / 0.95	99.95 / 5.02	99.95 / 5.01	99.94 / 4.93
CIC	S80	99.29 / 0.97	99.70 / 0.96	99.71 / 0.99	99.97 / 4.97	99.98 / 4.97	99.97 / 4.91
CIC	SQ	84.61 / 0.94	90.21 / 0.56	79.43 / 0.95	98.71 / 4.82	99.67 / 4.82	95.75 / 4.70
CIC	SQC	99.57 / 0.98	99.54 / 0.95	99.67 / 0.97	99.85 / 4.76	99.81 / 3.33	99.83 / 3.44
TON-hosts	F	98.17 / 4.77	98.12 / 0.00	98.32 / 4.77	99.95 / 5.64	99.88 / 0.91	99.92 / 5.60
TON-hosts	F8	98.18 / 0.00	97.96 / 0.00	98.29 / 0.00	99.70 / 0.28	99.82 / 1.10	99.81 / 5.16
TON-hosts	P	93.99 / 0.00	93.50 / 0.00	93.94 / 0.00	99.68 / 5.48	99.67 / 0.83	99.75 / 5.68
TON-hosts	Q	79.07 / 4.77	79.67 / 4.77	0.00 / 0.00	97.06 / 5.72	96.97 / 4.93	99.78 / 5.72
TON-hosts	QC	97.27 / 0.00	95.62 / 0.00	96.00 / 0.00	99.84 / 5.64	99.84 / 0.87	99.75 / 0.95
TON-hosts	R	96.85 / 0.00	95.86 / 0.00	97.01 / 0.00	99.80 / 4.61	99.84 / 0.91	99.80 / 0.91
TON-hosts	S80	96.95 / 0.00	97.10 / 0.00	95.40 / 0.00	99.72 / 5.75	99.83 / 5.12	99.86 / 0.75
TON-hosts	SQ	85.49 / 4.77	85.09 / 4.77	0.00 / 0.00	97.10 / 5.60	97.74 / 4.93	99.76 / 5.52
TON-hosts	SQC	96.92 / 0.00	96.85 / 0.00	94.12 / 0.00	99.75 / 4.65	99.80 / 4.45	99.82 / 0.83

Table 3: S2 (`S2_paired_differences.csv`): run-paired differences in recall and in achieved test FAR (points; seeds 1 2 3) between reported exports. The CSV also holds the record-level McNemar tests of job z9 with, for each, whether it was computed on the reported export or on a first-selected export that the per-tensor replacement superseded.

split	budget	pair	recall diff	test FAR diff
TON	1%	R - F	+2.41 +2.73 +0.35	+0.44 +0.44 +0.08
TON	1%	S80 - F	+1.68 +2.25 -1.26	+0.25 +0.27 -0.02
TON	1%	P - F	+1.99 -1.35 -0.95	+0.31 +0.29 -0.19
TON	1%	QC - F	+2.37 +1.91 -0.73	+0.23 +0.39 -0.12
TON	1%	R - P	+0.43 +4.08 +1.30	+0.12 +0.14 +0.27
TON	1%	S80 - P	-0.31 +3.60 -0.32	-0.06 -0.02 +0.17
TON	1%	QC - P	+0.38 +3.26 +0.22	-0.08 +0.10 +0.06
TON	1%	R - S80	+0.73 +0.48 +1.62	+0.19 +0.17 +0.10
TON	1%	R - QC	+0.04 +0.82 +1.08	+0.21 +0.04 +0.21
TON	1%	SQC - S80	-1.22 +0.27 +1.39	-0.02 +0.04 -0.02
TON	5%	R - F	-0.03 -0.04 -0.08	-0.35 +0.27 -0.37
TON	5%	S80 - F	-0.04 -0.08 +0.01	-0.10 +0.33 +0.06
TON	5%	P - F	-0.09 -0.12 -0.15	-0.23 +0.08 -0.02
TON	5%	QC - F	-0.03 -0.04 -0.05	-0.23 +0.19 +0.08
TON	5%	R - P	+0.06 +0.08 +0.07	-0.12 +0.19 -0.35
TON	5%	S80 - P	+0.05 +0.04 +0.16	+0.12 +0.25 +0.08
TON	5%	QC - P	+0.06 +0.08 +0.10	+0.00 +0.10 +0.10
TON	5%	R - S80	+0.01 +0.04 -0.09	-0.25 -0.06 -0.44
TON	5%	R - QC	+0.00 +0.00 -0.03	-0.12 +0.08 -0.46
TON	5%	SQC - S80	-0.04 +0.03 -0.03	-0.73 -0.33 +0.08
CIC	1%	R - F	+1.88 +4.89 +1.36	-0.03 -0.14 -0.05
CIC	1%	S80 - F	+1.47 +5.17 +1.39	-0.01 -0.02 -0.01
CIC	1%	P - F	+1.05 +4.41 +1.23	+0.00 -0.17 -0.04
CIC	1%	QC - F	+1.80 +4.91 +1.31	-0.05 -0.06 -0.04
CIC	1%	R - P	+0.83 +0.48 +0.12	-0.03 +0.03 -0.01
CIC	1%	S80 - P	+0.42 +0.75 +0.15	-0.02 +0.15 +0.03
CIC	1%	QC - P	+0.75 +0.50 +0.07	-0.05 +0.11 +0.00

split	budget	pair	recall diff	test FAR diff
CIC	1%	R - S80	+0.41 -0.27 -0.03	-0.01 -0.12 -0.04
CIC	1%	R - QC	+0.08 -0.02 +0.05	+0.02 -0.07 -0.01
CIC	1%	SQC - S80	+0.27 -0.16 -0.04	+0.01 -0.01 -0.02
CIC	5%	R - F	-0.02 +0.23 -0.04	-0.09 -0.05 -0.10
CIC	5%	S80 - F	-0.01 +0.26 -0.01	-0.14 -0.10 -0.12
CIC	5%	P - F	-0.13 +0.12 -0.24	-0.54 -0.27 -0.73
CIC	5%	QC - F	-0.18 +0.10 -0.18	-1.12 -0.83 -1.87
CIC	5%	R - P	+0.11 +0.11 +0.20	+0.45 +0.21 +0.63
CIC	5%	S80 - P	+0.13 +0.14 +0.23	+0.40 +0.17 +0.61
CIC	5%	QC - P	-0.04 -0.01 +0.05	-0.58 -0.56 -1.14
CIC	5%	R - S80	-0.02 -0.03 -0.03	+0.06 +0.05 +0.02
CIC	5%	R - QC	+0.15 +0.13 +0.14	+1.04 +0.78 +1.77
CIC	5%	SQC - S80	-0.12 -0.17 -0.14	-0.21 -1.64 -1.48
TON-hosts	1%	R - F	-1.32 -2.25 -1.31	-4.77 +0.00 -4.77
TON-hosts	1%	S80 - F	-1.23 -1.01 -2.93	-4.77 +0.00 -4.77
TON-hosts	1%	P - F	-4.18 -4.62 -4.38	-4.77 +0.00 -4.77
TON-hosts	1%	QC - F	-0.90 -2.50 -2.32	-4.77 +0.00 -4.77
TON-hosts	1%	R - P	+2.86 +2.37 +3.07	+0.00 +0.00 +0.00
TON-hosts	1%	S80 - P	+2.96 +3.61 +1.45	+0.00 +0.00 +0.00
TON-hosts	1%	QC - P	+3.29 +2.12 +2.06	+0.00 +0.00 +0.00
TON-hosts	1%	R - S80	-0.10 -1.24 +1.62	+0.00 +0.00 +0.00
TON-hosts	1%	R - QC	-0.43 +0.25 +1.01	+0.00 +0.00 +0.00
TON-hosts	1%	SQC - S80	-0.03 -0.25 -1.28	+0.00 +0.00 +0.00
TON-hosts	5%	R - F	-0.14 -0.03 -0.12	-1.02 +0.00 -4.69
TON-hosts	5%	S80 - F	-0.22 -0.04 -0.06	+0.12 +4.22 -4.85
TON-hosts	5%	P - F	-0.27 -0.21 -0.17	-0.16 -0.08 +0.08
TON-hosts	5%	QC - F	-0.10 -0.03 -0.17	+0.00 -0.04 -4.65
TON-hosts	5%	R - P	+0.13 +0.18 +0.04	-0.87 +0.08 -4.77
TON-hosts	5%	S80 - P	+0.05 +0.17 +0.11	+0.28 +4.30 -4.93
TON-hosts	5%	QC - P	+0.17 +0.18 +0.00	+0.16 +0.04 -4.73
TON-hosts	5%	R - S80	+0.08 +0.01 -0.06	-1.14 -4.22 +0.16
TON-hosts	5%	R - QC	-0.04 +0.00 +0.04	-1.02 +0.04 -0.04
TON-hosts	5%	SQC - S80	+0.03 -0.03 -0.04	-1.10 -0.67 +0.08

Table 4: S3 (`S3_device_disagreements.csv`): device decisions vs. the host TFLite interpreter on the same file, at the frozen thresholds; the CSV splits every count by input category (random test records, near-threshold records, records beyond 5 SD, synthetic extremes, zero vectors).

board	model	weights	act.	FAR	agree	add. misses	add. detections	added FA	avoided FA	probes
ESP32 (TFLM 1.0.0)	ton_R80	per-tensor	int8	0.01	4462/4462	0	0	0	0	0
ESP32 (TFLM 1.0.0)	ton_R80	per-tensor	int8	0.05	4461/4462	0	0	0	1	0
ESP32-S3 (TFLM 1.0.0)	ton_R80	per-tensor	int8	0.01	4462/4462	0	0	0	0	0
ESP32-S3 (TFLM 1.0.0)	ton_R80	per-tensor	int8	0.05	4461/4462	0	0	0	1	0
ESP32 (TFLM 1.0.0)	ton_S80	per-channel	int8	0.01	2209/4462	2069	0	0	23	161
ESP32 (TFLM 1.0.0)	ton_S80	per-channel	int8	0.05	4269/4462	49	0	0	89	55
ESP32-S3 (TFLM 1.0.0)	ton_S80	per-channel	int8	0.01	2209/4462	2069	0	0	23	161
ESP32-S3 (TFLM 1.0.0)	ton_S80	per-channel	int8	0.05	4269/4462	49	0	0	89	55
ESP32 (TFLM 1.0.0)	ton_P80	per-channel	int8	0.01	3141/4462	929	223	2	10	157
ESP32 (TFLM 1.0.0)	ton_P80	per-channel	int8	0.05	4369/4462	24	0	0	12	57
ESP32-S3 (TFLM 1.0.0)	ton_P80	per-channel	int8	0.01	3141/4462	929	223	2	10	157
ESP32-S3 (TFLM 1.0.0)	ton_P80	per-channel	int8	0.05	4369/4462	24	0	0	12	57
ESP32 (TFLM 1.0.0)	ton_QC80	per-tensor	int8	0.01	4456/4462	0	0	6	0	0
ESP32 (TFLM 1.0.0)	ton_QC80	per-tensor	int8	0.05	4462/4462	0	0	0	0	0
ESP32-S3 (TFLM 1.0.0)	ton_QC80	per-tensor	int8	0.01	4456/4462	0	0	6	0	0
ESP32-S3 (TFLM 1.0.0)	ton_QC80	per-tensor	int8	0.05	4462/4462	0	0	0	0	0
ESP32 (TFLM 1.0.0)	ton_PQ80_minmax	per-tensor	int8	0.01	4161/4462	0	7	294	0	0
ESP32 (TFLM 1.0.0)	ton_PQ80_minmax	per-tensor	int8	0.05	4134/4462	0	32	295	0	1
ESP32-S3 (TFLM 1.0.0)	ton_PQ80_minmax	per-tensor	int8	0.01	4161/4462	0	7	294	0	0
ESP32-S3 (TFLM 1.0.0)	ton_PQ80_minmax	per-tensor	int8	0.05	4134/4462	0	32	295	0	1
ESP32 (TFLM 1.0.0)	ton_PQ80_int16	per-channel	int16	0.01	3379/4462	933	4	0	12	134
ESP32 (TFLM 1.0.0)	ton_PQ80_int16	per-channel	int16	0.05	4346/4462	23	0	0	26	67
ESP32-S3 (TFLM 1.0.0)	ton_PQ80_int16	per-channel	int16	0.01	3379/4462	933	4	0	12	134
ESP32-S3 (TFLM 1.0.0)	ton_PQ80_int16	per-channel	int16	0.05	4346/4462	23	0	0	26	67
ESP32 (TFLM 1.0.0)	ton_S80_pt	per-tensor	int8	0.01	4462/4462	0	0	0	0	0
ESP32 (TFLM 1.0.0)	ton_S80_pt	per-tensor	int8	0.05	4458/4462	0	0	0	4	0
ESP32-S3 (TFLM 1.0.0)	ton_S80_pt	per-tensor	int8	0.01	4462/4462	0	0	0	0	0
ESP32-S3 (TFLM 1.0.0)	ton_S80_pt	per-tensor	int8	0.05	4458/4462	0	0	0	4	0

board	model	weights	act.	FAR	agree	add. misses	add. detections	added FA	avoided FA	probes
ESP32 (TFLM 1.0.0)	ton_P80_pt	per-tensor	int8	0.01	4461/4462	0	0	0	0	1
ESP32 (TFLM 1.0.0)	ton_P80_pt	per-tensor	int8	0.05	4462/4462	0	0	0	0	0
ESP32-S3 (TFLM 1.0.0)	ton_P80_pt	per-tensor	int8	0.01	4461/4462	0	0	0	0	1
ESP32-S3 (TFLM 1.0.0)	ton_P80_pt	per-tensor	int8	0.05	4462/4462	0	0	0	0	0
ESP32 (TFLM 1.0.0)	cic_R80	per-tensor	int8	0.01	5882/5902	0	0	11	9	0
ESP32 (TFLM 1.0.0)	cic_R80	per-tensor	int8	0.05	5885/5902	0	0	13	4	0
ESP32-S3 (TFLM 1.0.0)	cic_R80	per-tensor	int8	0.01	5882/5902	0	0	11	9	0
ESP32-S3 (TFLM 1.0.0)	cic_R80	per-tensor	int8	0.05	5885/5902	0	0	13	4	0
ESP32 (TFLM 1.0.0)	cic_S80	per-tensor	int8	0.01	5898/5902	1	0	3	0	0
ESP32 (TFLM 1.0.0)	cic_S80	per-tensor	int8	0.05	5880/5902	0	0	7	15	0
ESP32-S3 (TFLM 1.0.0)	cic_S80	per-tensor	int8	0.01	5898/5902	1	0	3	0	0
ESP32-S3 (TFLM 1.0.0)	cic_S80	per-tensor	int8	0.05	5880/5902	0	0	7	15	0
ESP32 (TFLM 1.0.0)	cic_P80	per-tensor	int8	0.01	5878/5902	0	0	23	1	0
ESP32 (TFLM 1.0.0)	cic_P80	per-tensor	int8	0.05	5877/5902	0	0	19	6	0
ESP32-S3 (TFLM 1.0.0)	cic_P80	per-tensor	int8	0.01	5878/5902	0	0	23	1	0
ESP32-S3 (TFLM 1.0.0)	cic_P80	per-tensor	int8	0.05	5877/5902	0	0	19	6	0
ESP32 (TFLM 1.0.0)	cic_QC80	per-tensor	int8	0.01	5511/5902	0	8	382	1	0
ESP32 (TFLM 1.0.0)	cic_QC80	per-tensor	int8	0.05	5900/5902	0	0	1	1	0
ESP32-S3 (TFLM 1.0.0)	cic_QC80	per-tensor	int8	0.01	5511/5902	0	8	382	1	0
ESP32-S3 (TFLM 1.0.0)	cic_QC80	per-tensor	int8	0.05	5900/5902	0	0	1	1	0
ESP32 (TFLM 1.0.0)	cic_PQ80_minmax	per-tensor	int8	0.01	5845/5902	0	7	49	0	1
ESP32 (TFLM 1.0.0)	cic_PQ80_minmax	per-tensor	int8	0.05	5895/5902	0	0	0	7	0
ESP32-S3 (TFLM 1.0.0)	cic_PQ80_minmax	per-tensor	int8	0.01	5845/5902	0	7	49	0	1
ESP32-S3 (TFLM 1.0.0)	cic_PQ80_minmax	per-tensor	int8	0.05	5895/5902	0	0	0	7	0
ESP32 (TFLM 1.0.0)	cic_PQ80_int16	per-channel	int16	0.01	4952/5902	82	24	693	136	15
ESP32 (TFLM 1.0.0)	cic_PQ80_int16	per-channel	int16	0.05	5292/5902	38	4	139	400	29
ESP32-S3 (TFLM 1.0.0)	cic_PQ80_int16	per-channel	int16	0.01	4952/5902	82	24	693	136	15
ESP32-S3 (TFLM 1.0.0)	cic_PQ80_int16	per-channel	int16	0.05	5292/5902	38	4	139	400	29

Table 5: S5 (`S5_stream_completeness.csv`): stream completeness. Final job, outputs present, device resets and skipped inputs in the final run, bit-identity with the other board, earlier incomplete runs (superseded). Duplicates are impossible: the protocol is lockstep, one reply per input index.

board	model	job	outputs	resets	skipped	bit-identical	earlier runs
ESP32 (TFLM 1.0.0)	ton_R80	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_R80	z8	4462/4462	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 4462/4462	
ESP32 (TFLM 1.0.0)	ton_S80	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_S80	z8	4462/4462	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 4462/4462	
ESP32 (TFLM 1.0.0)	ton_P80	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_P80	z8	4462/4462	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 4462/4462	
ESP32 (TFLM 1.0.0)	ton_QC80	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_QC80	z8c	4462/4462	0	0	ESP32 (TFLM 1.0.0): 4462/4462	2026-11-05_z8_device_e7: n=None, crash at None, error=ERROR no reply t
ESP32 (TFLM 1.0.0)	ton_PQ80_minmax	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_PQ80_minmax	z8	4462/4462	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 4462/4462	
ESP32 (TFLM 1.0.0)	ton_PQ80_int16	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_PQ80_int16	z8	4462/4462	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 4462/4462	
ESP32 (TFLM 1.0.0)	ton_S80_pt	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_S80_pt	z8c	4462/4462	0	0	ESP32 (TFLM 1.0.0): 4462/4462	2026-11-05_z8_device_e7: n=601, crash at 601, error=None
ESP32 (TFLM 1.0.0)	ton_P80_pt	z8b	4462/4462	0	0	ESP32-S3 (TFLM 1.0.0): 4462/4462	
ESP32-S3 (TFLM 1.0.0)	ton_P80_pt	z8c	4462/4462	0	0	ESP32 (TFLM 1.0.0): 4462/4462	2026-11-05_z8_device_e7: n=None, crash at None, error=ERROR no reply t
ESP32 (TFLM 1.0.0)	cic_R80	z8b	5902/5902	0	0	ESP32-S3 (TFLM 1.0.0): 5902/5902	
ESP32-S3 (TFLM 1.0.0)	cic_R80	z8c	5902/5902	1	0	ESP32 (TFLM 1.0.0): 5902/5902	2026-11-05_z8_device_e7: n=None, crash at None, error=ERROR no reply t
ESP32 (TFLM 1.0.0)	cic_S80	z8b	5902/5902	0	0	ESP32-S3 (TFLM 1.0.0): 5902/5902	
ESP32-S3 (TFLM 1.0.0)	cic_S80	z8	5902/5902	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 5902/5902	
ESP32 (TFLM 1.0.0)	cic_P80	z8b	5902/5902	0	0	ESP32-S3 (TFLM 1.0.0): 5902/5902	
ESP32-S3 (TFLM 1.0.0)	cic_P80	z8c	5902/5902	1	0	ESP32 (TFLM 1.0.0): 5902/5902	2026-11-05_z8_device_e7: n=2559, crash at 2559, error=None
ESP32 (TFLM 1.0.0)	cic_QC80	z8b	5902/5902	0	0	ESP32-S3 (TFLM 1.0.0): 5902/5902	
ESP32-S3 (TFLM 1.0.0)	cic_QC80	z8	5902/5902	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 5902/5902	
ESP32 (TFLM 1.0.0)	cic_PQ80_minmax	z8b	5902/5902	0	0	ESP32-S3 (TFLM 1.0.0): 5902/5902	
ESP32-S3 (TFLM 1.0.0)	cic_PQ80_minmax	z8	5902/5902	not logged (z8 host loop)	0	ESP32 (TFLM 1.0.0): 5902/5902	
ESP32 (TFLM 1.0.0)	cic_PQ80_int16	z8b	5902/5902	0	0	ESP32-S3 (TFLM 1.0.0): 5902/5902	
ESP32-S3 (TFLM 1.0.0)	cic_PQ80_int16	z8c	5902/5902	1	0	ESP32 (TFLM 1.0.0): 5902/5902	2026-11-05_z8_device_e7: n=4137, crash at 4137, error=None

Table 6: S4 (`S4_firmware_memory.csv`, `S4_model_memory_latency.csv`): firmware memory from the linker (bytes). Static RAM includes the 49,152-byte tensor arena; each model uses 1.4–1.8 KB of it on TFLM 1.0.0 and 1.9–5.9 KB on the current runtime (per-model file). Heap was logged only in the z10 builds; it does not change during the stream. Latency is measured around `Invoke()` only (the graph's QUANTIZE, four FULLY_CONNECTED, LOGISTIC, DEQUANTIZE); copying the features into the input tensor is reported separately and serial transfer is excluded.

chip	runtime	models	static RAM	arena	flash	models	heap free	min free
ESP32-S3	TFLM 1.0.0 (Arduino TensorFlowLite_ESP32)	14	74828	49152	1409625	915336		
ESP32	TFLM 1.0.0 (Arduino TensorFlowLite_ESP32)	14	77612	49152	1418881	915336		
ESP32	TFLM 1.0.0, rebuilt for z10 (heap logged)	14	77612	49152	1419293	915336	291168	285568
ESP32	esp-tflite-micro 1.4.2 / ESP-IDF 5.5, esp-nn	3	61364	49152	440177	198576	215024	215024
ESP32	esp-tflite-micro 1.4.2 / ESP-IDF 5.5, reference kernels	3	61364	49152	440177	198576	215024	215024

S6: runtime comparison

See `docs/runtime_compare/runtime_compare.{md,csv}` (same three files and 4,462 TON_IoT inputs on the same classic ESP32; runtime-comparison table of the paper). Build provenance (component hashes, `dependencies.lock`, `sdkconfig`) is in each `device_e7/<board>/build_meta/` of jobs z10, z10d, z10f.

Table 7: S7a (`S7_split_provenance.csv`): overlap of the revised splits with the earlier (exploratory) split. Test records were all in the earlier training pool, which only exploratory models of the earlier pipeline used; none was in the earlier test split. No reported model, threshold or calibration draw used a test record.

split	partition	records	in earlier test	in earlier training pool
ton_random	new_test	21032	0	21032
ton_random	new_client_train	75716	18927	56789
ton_random	new_client_val	8412	2105	6307
ton_hosts	new_test	22866	0	22866
ton_hosts	new_client_train	68937	13797	55140
ton_hosts	new_client_val	7660	1538	6122
cic_random	new_test	499038	0	499038
cic_random	new_client_train	1529825	382551	1147274
cic_random	new_client_val	199616	49739	149877

Table 8: S7b: method-freeze history (UTC, from the repository history).

time	event
2026-10-28 02:00–02:14	Pipeline v13, training recipe, arms F/F8/P/Q/R/S/SQ, budgets (80/128 KB; FAR 1%/5%), threshold rule and strong-PTQ candidates committed, before any v13 result.
2026-10-29 05:48	v13.1 rules committed (drop impossible negative values; the new test split excludes every record of the earlier test split), before any v13 test-split result was used.
2026-10-29 07:24	Float32 threshold fix (validation-side assertion); the two affected jobs were re-run (z1d, z1e) with unchanged models.
2026-10-29 07:29	E3/E4 design committed before any E3/E4 result.
2026-10-29 15:58	*Post hoc*, after E1–E4 test results: QC/SQC arms, E4 clipped-FP32 and 16-bit rows, E6 host split.
2026-11-05 00:57	*Post hoc*, after device results: per-tensor re-exports of the TON S80/P80 device models.
2026-11-06	*Post hoc*: per-tensor strong-PTQ replacements for every F8/P/S80 run (z9, same calibration draw and selection rule); runtime comparison (z10, z10d, z10f). No model retrained, no threshold changed.